Soccer Markov Nash · research note · 15 October 2026

Policy-gradient experiment assumptions

A record of the choices behind the additional comparisons, how outcomes were evaluated, and safeguards against reward hacking.

Purpose. These runs compare training procedures with the repository's existing game and exact finite-horizon evaluator. They do not establish convergence, a new algorithmic guarantee, or publication readiness. Values below are exploratory evidence and should not be presented as confirmatory results.

Reward and reporting safeguards

- **The game reward and transition rules stay fixed within each comparison.** No reward shaping, win-rate bonus, or altered terminal reward was introduced to raise the reported score.

- **Training entropy is a separate optimization setting.** The default coefficient is 0.01; a 0.05 run is reported as an explicit ablation. It changes the policy objective's regularization term, not the soccer payoff or evaluator.

- **These are exploratory runs, not a confirmatory sweep.** Report every tested setting and seed; before a new comparison, freeze the game, reward, hyperparameters, seeds, and primary metric, then evaluate on fresh seeds. Win rate is diagnostic only: the exact best response optimizes discounted score difference, not win probability.

Environment assumptions

Choice	Setting used	Consequence
Discount and horizon	$\gamma = 0.9$; 100 steps maximum	The networks receive state and remaining horizon. The exact reference solves the matching finite-horizon game.
First-goal format	Episode ends when either side scores	Outcome is which side scores first; deterministic A10 and random move-order boards are reported separately.
Goal-reset format	After a goal, reset play and continue to the 100-step limit	Outcome is score difference over the full horizon; tested here on deterministic A10. It is not pooled with first-goal results.
Board and rules	Repository A10 deterministic rules or random move-order rules	"Random board" means stochastic move order in this project, not a newly sampled board layout.
Seat handling	Player 0 and player 1 evaluated separately	Do not average seats without stating weights. Exact-solver win rates vs. exact best response on the random move-order board: 67.2% for player 0, 39.3% for player 1.

Training and evaluation assumptions

- The earlier PG benchmark uses seeds 0–2. The new matched A10 pilot uses seeds 4–6, 2,000 iterations, 64 episodes per batch, five gradient passes per collected batch, and entropy coefficient 0.01. Its standard A2C context arm uses the usual 10-step bootstrapped advantage. The learned-critic MC control and frozen exact-value critic both use sampled full returns for the actor; only the state-value baseline training differs. Seed IDs do not create common random-number trajectories. Players have independent policy/value networks and no imposed mirror symmetry.

- Policies are snapshotted every five iterations. Training samples an opponent snapshot uniformly; the reported policy averages its action probabilities state by state. Greedy snapshot selection and network-weight averaging are separate ablations, not interchangeable forms of fictitious play.

- Each seat is evaluated in 1,000 kickoff games. Exact finite-horizon best responses produce exploitability in discounted-value units; the best response optimizes that value, not win chance. Win/tie/loss against random, exact Nash, and exact best response are separate sampled frequencies.

- The frozen-value experiment fits each critic to exact state-time values once, freezes it, and updates the actor with sampled discounted returns minus that baseline. It does not query exact action values during actor updates. The separate `a2c_exact` method uses oracle advantages and is a different experiment. The matched `a2c_mc` control learns its baseline from the same sampled full returns used by the actor estimator.

Observed exploratory comparisons

What these runs support

Mean results are shown with the full seed range. The table separates game formats and training interventions; the final three rows are the new paired A10 pilot.

Condition	Seeds	Exploitability	Player 0 wins vs exact BR	Reading
A10, first-goal, 1 update	0–2	0.888 mean	0.63% mean	Reference training setting; not near exact equilibrium.
A10, first-goal, 5 updates	0–2	0.900 mean	0.10% mean	No improvement over one update.
A10, first-goal, greedy snapshot response	0–2	0.884 mean	1.40% mean	Small win-rate change; still loses almost every game to exact BR.
A10, first-goal, entropy 0.05	0–2	0.570 mean	0.23% mean	Lower exploitability in this ablation, with negligible BR win rate; not convergence.
A10, first-goal, 1 update	3 check	0.889	0.9%	Fresh-seed check after exploratory results; player 1 won 2.2% vs its exact BR.
A10, first-goal, 5 updates	3 check	0.928	0.8%	Did not improve on the matching one-update check; player 1 won 1.1% vs its exact BR.
A10, first-goal, entropy 0.05	3 check	0.570	0.1%	Lower value gap persisted; player 1 won 0.2% vs its exact BR.
A10, first-goal, frozen exact-value baseline	4–6	0.878 mean (0.808–0.928)	0.7% mean	Player 1 won 0.7% on average. Paired with the learned-critic MC control; direction varies across seeds.
A10, first-goal, learned-critic MC control	4–6	0.865 mean (0.793–0.940)	3.0% mean	Same full-return actor estimator and training budget as the frozen-value arm; learned baseline.
A10, first-goal, standard A2C context	4–6	0.877 mean (0.845–0.898)	1.7% mean	Uses the usual 10-step estimator, so it is not the isolated critic-pretraining control.
A10, first-goal, policy vs. weight averaging	0 only	0.937 vs. 0.980 / 0.962	0.10% / 0.10% / 0%	Policy average / averaged weights / last weights; one seed only.
Random move order, first-goal, 1 update	0–2	0.640 mean	12.3% mean	Seat-sensitive stochastic-game comparison.
Random move order, first-goal, 5 updates	0–2	0.589 mean	14.8% mean	Seed 0 accounts for much of the apparent gain; seed 1/2 results are weaker.
A10, continuing goal-reset, 1 vs 5 updates	0 only	1.567 vs 1.010	0% both	Exploitability fell in this single seed; no multi-seed confirmation.

The seed-3 checks were selected after inspecting seeds 0–2 and therefore are not a preregistered holdout. The seeds 4–6 pilot was a later fixed-seed comparison, not a preregistered confirmatory study. Raw outputs are in `experiments/pg_compare_*.csv`.

Incomplete run. A second continuing-format seed was stopped before completion at the requested time cutoff. It produced no result and is excluded. Do not infer that the single-seed change generalizes.

Interpretation and next steps

Limits and next comparison

- None of these learned policies matches the exact equilibrium. In the deterministic A10 runs, wins against the value-best response stay near zero. The random move-order results also fall below the exact solver's seat-specific baseline in important matchups.

- In the seeds 4–6 matched pilot, the frozen-minus-learned-MC Nash-gap differences were +0.066, –0.041, and +0.015 (mean +0.013). This sample is too small and inconsistent to support a reliable benefit or harm claim.

- The entropy ablation is exploratory and was inspected against the same evaluation metrics. Seed 3 repeats the lower exploitability but not meaningful best-response wins. It must remain labeled an ablation; the current evidence is not an unbiased test of a preselected method.

- Before making a research claim, fix a reward and evaluator, specify both game formats, preselect one primary value-based metric and the seat aggregation, use fresh seeds, report uncertainty, and publish all configurations and failures.

- The directly related soccer study, Competitive Policy Optimization (UAI 2021), evaluates a different game/reward and a different algorithm. Its results motivate future algorithmic comparisons; they do not validate these runs.

Values are rounded from the script's CSV output. Sampled win rates are Monte Carlo estimates from 1,000 games per seat; exploitability uses exact responses and is a different metric. Implementation: `scripts/pg_finite.py`, `soccer_nash/pg_finite.py`, `soccer_nash/finite_horizon.py`. Broader game and solver assumptions: `solver_assumptions.md` and `assumptions.md`.